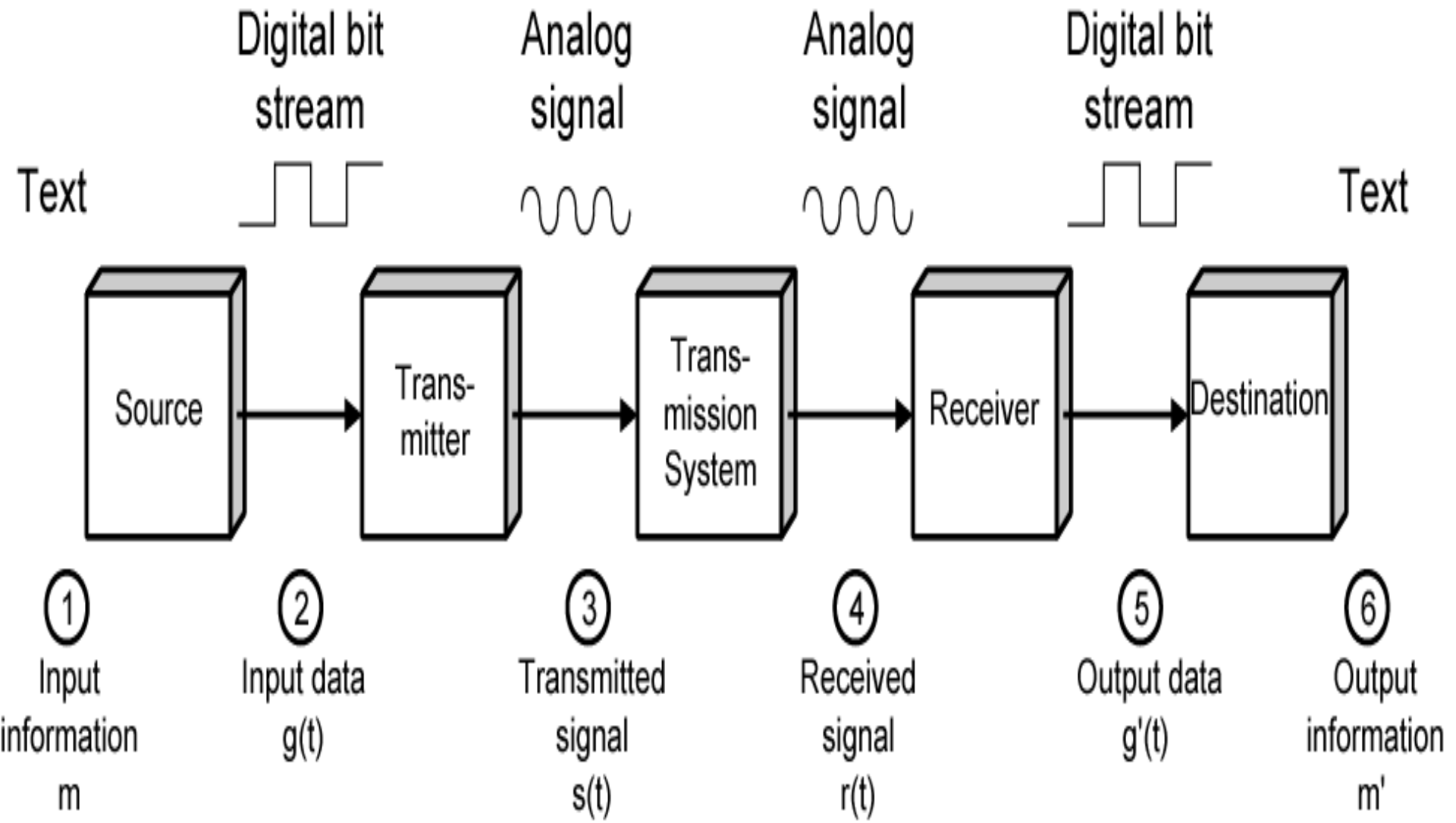
Selamat Pagi
saudara-saudara

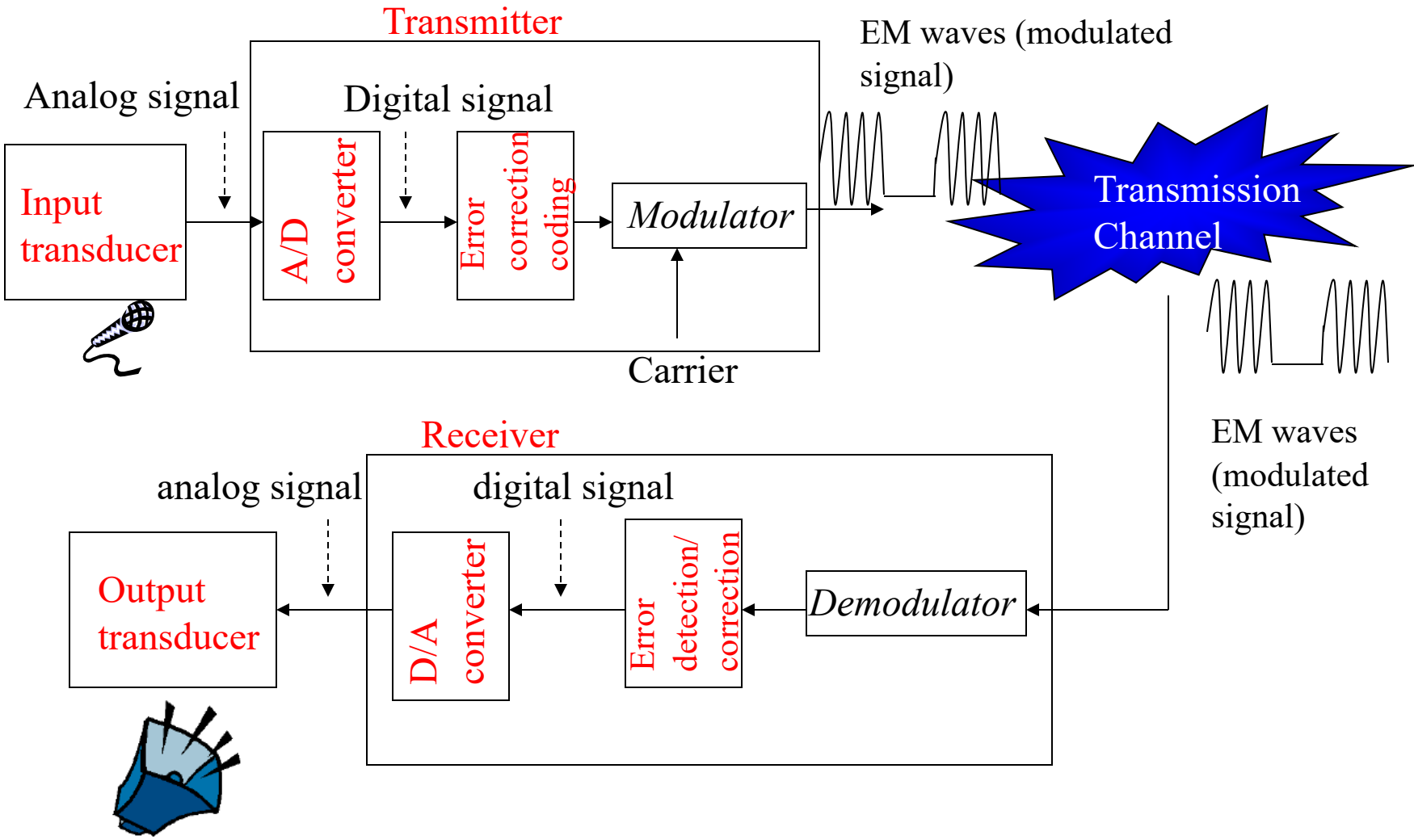
Analog Transmission of Digital Data:

ASK, FSK, PSK, QAM

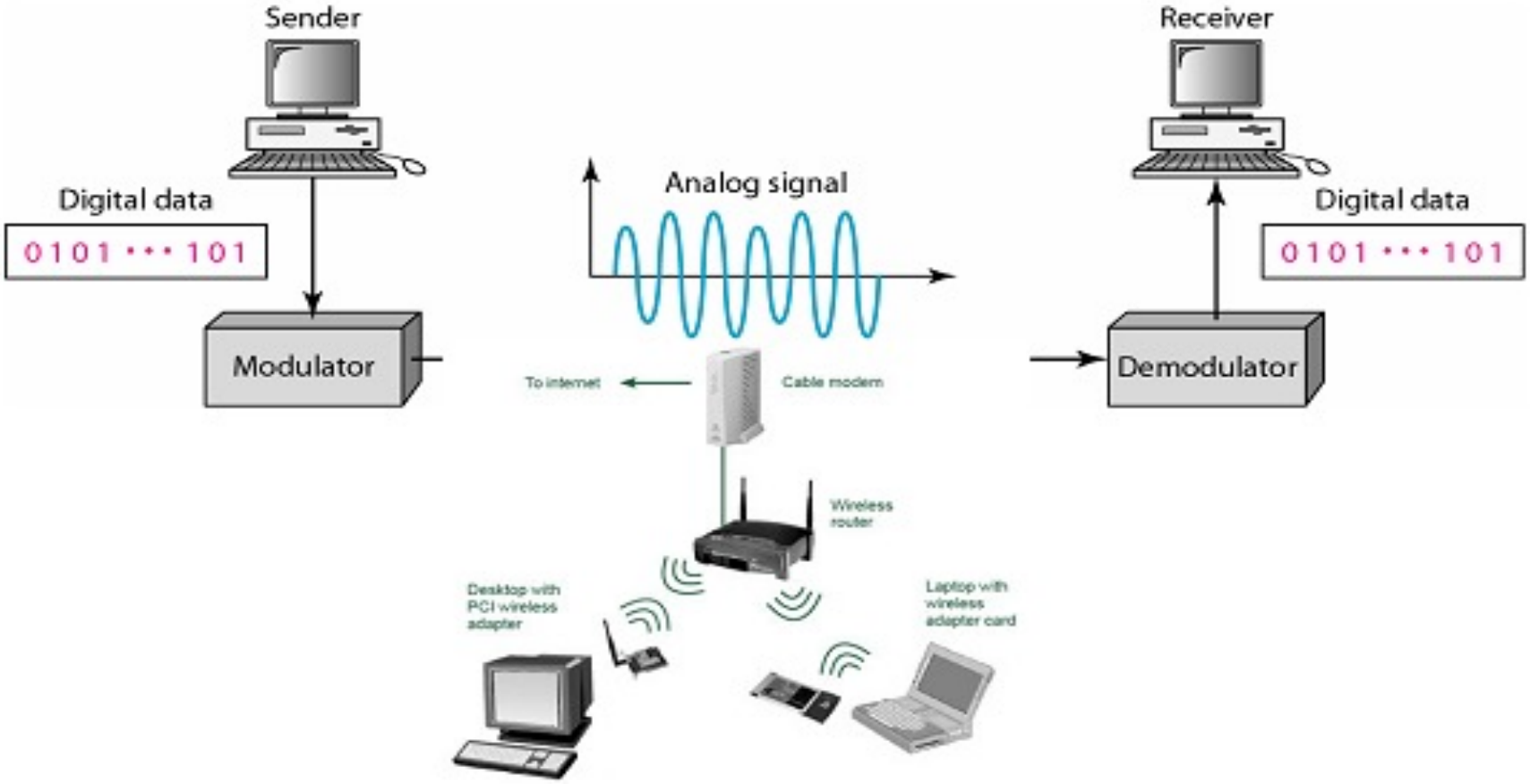
- Digital communication systems also employ modulation techniques, some of which include:
  - Amplitude Shift Keying (ASK)
  - Frequency Shift Keying (FSK)
  - Phase Shift Keying (PSK)
  - Quadrature Amplitude Modulation (QAM)


Digital bit stream
Analog signal
Analog signal
Digital bit stream
Text
Text
Source
Trans-mitter
Trans-mission System
Receiver
Destination
1
2
3
4
5
6
Input information m
Input data g(t)
Transmitted signal s(t)
Received signal r(t)
Output data g'(t)
Output information m'

Basic digital communications system



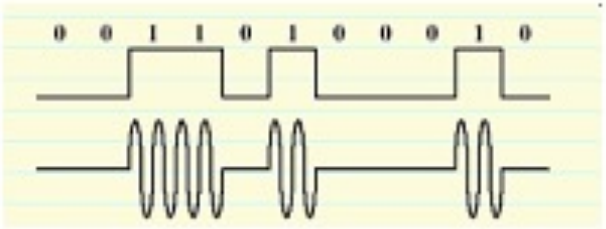
Why Do We Need Digital-to-Analog Conversion?!



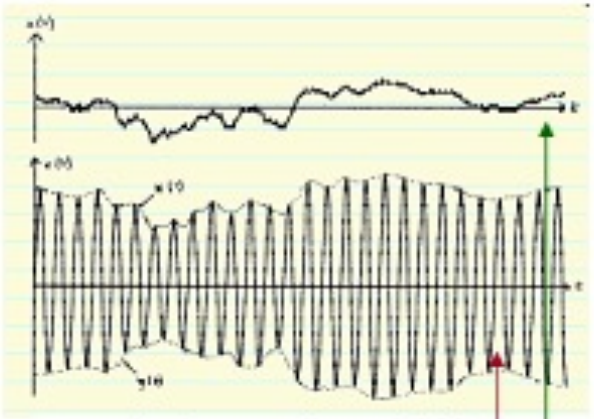
1) The medium/channel is band pass, and/or
2) Multiple users need to share the medium.

Modulation of Digital Data

Modulation – process of converting digital data or a low-pass analog to band-pass (higher-frequency) analog signal



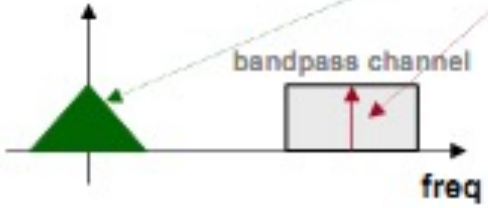
Digital-to-analog modulation.



Analog-to-analog modulation.

Carrier Signal – aka carrier freq. or modulated signal - high freq. signal that acts as a basis for the information signal

- information signal is called modulating signal

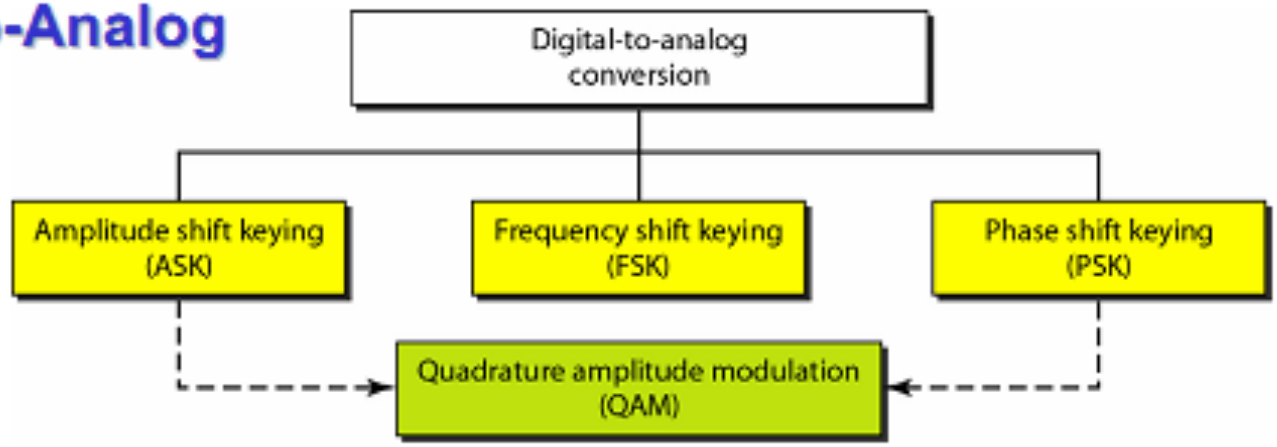


Modulation of Digital Data (cont.)

Digital-to-Analog Modulation – process of changing one of the characteristic of an analog signal (typically a sinewave) based on the information in a digital signal

- sinewave is defined by 3 characteristics (amplitude, frequency, and phase) ⇒ digital data (binary 0 & 1) can be represented by varying any of the three
- application: transmission of digital data over telephone wire (modem)

Types of Digital-to-Analog Modulation



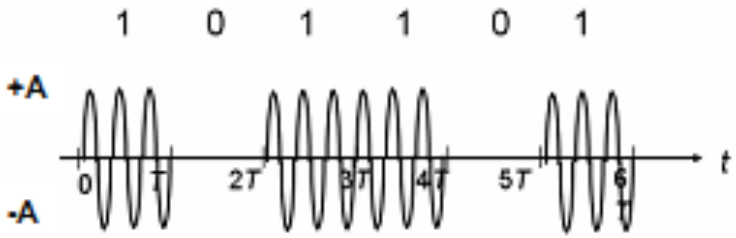
Modulation of Digital Data: ASK

ASK – strength of carrier signal is varied to represent binary 1 or 0

- both frequency & phase remain constant while amplitude changes
- commonly, one of the amplitudes is zero

$$s(t) = \begin{cases} A_0 \cos(2\pi f_c t), & \text{binary 0} \\ A_1 \cos(2\pi f_c t), & \text{binary 1} \end{cases}$$

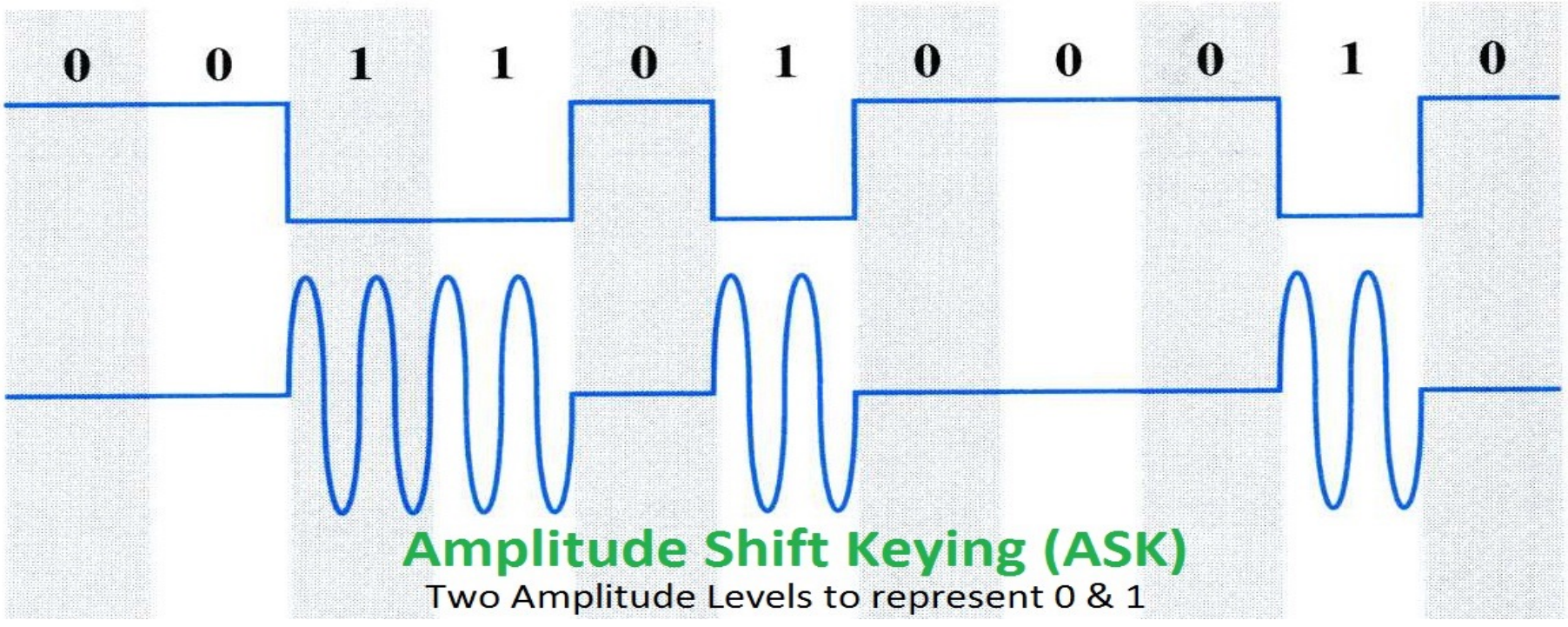
Is this picture, from the textbook, entirely correct?!



- demodulation: only the presence or absence of a sinusoid in a given time interval needs to be determined
- advantage: simplicity
- disadvantage: ASK is very susceptible to noise interference – noise usually (only) affects the amplitude, therefore ASK is the modulation technique most affected by noise
- application: ASK is used to transmit digital data over optical fiber

Amplitude Shift Keying

- Amplitude Shift Keying (ASK)
  - The most basic (binary) form of ASK involves the process of switching the carrier either on or off, in correspondence to a sequence of digital pulses that constitute the information signal. One binary digit is represented by the presence of a carrier, the other binary digit is represented by the absence of a carrier. Frequency remains fixed

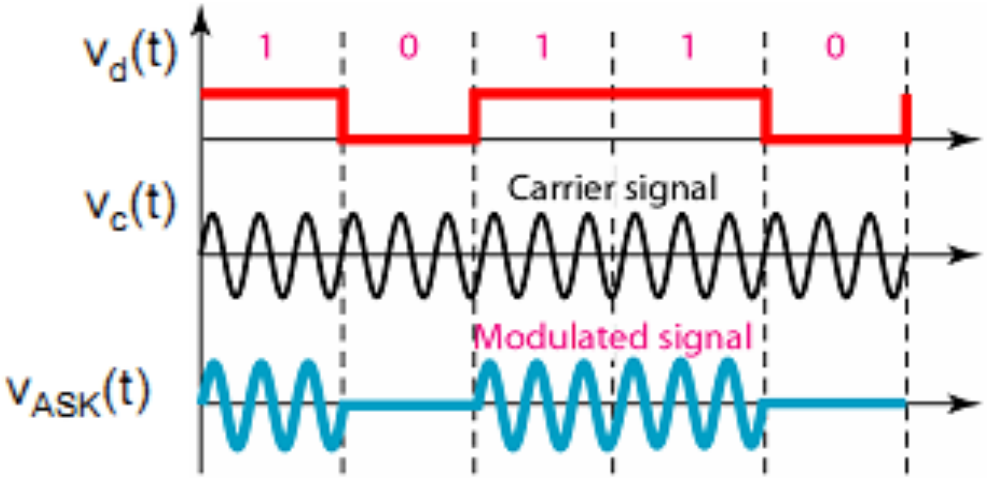


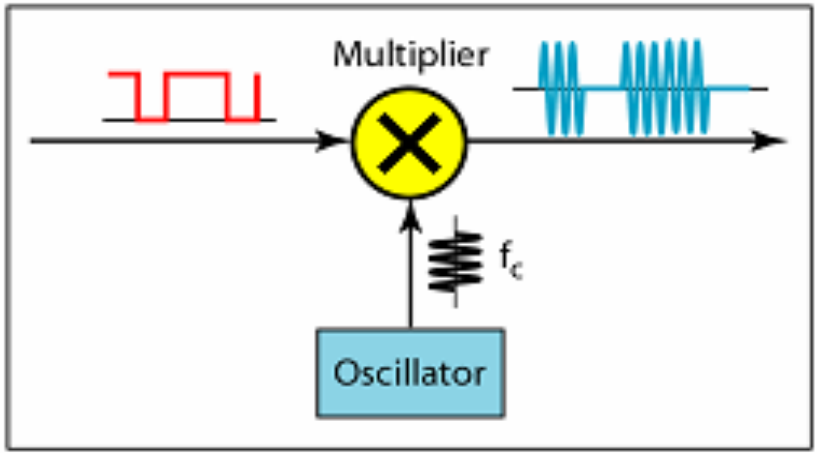
- Weakness of ASK: Susceptible to sudden noise impulses (lightning).
  ASK cannot be used for high speed transmission.
- ASK: Amplitude Shift Keying typically transmit data up to 1200 bps on voice-grade lines (dialup modems).

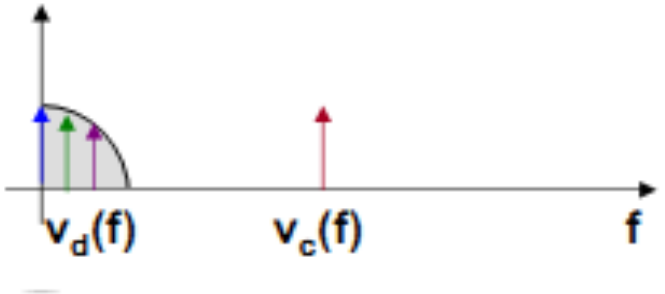
- Hasil diwakili oleh perbedaan amplitudo pada carrier Selalu, satu amplitudo adalah zero
  Yakni,kehadiran dan ketidakhadiran pada carrier adalah digunakan
- Rentan untuk pergantian gain tiba-tiba
- Tidak efisien
- Sampai dengan 1200bps pada voice grade line
- Digunakan pada fiber optic

Modulation of Digital Data: ASK (cont.)

Example [ASK]







How does the frequency spectrum of $v_{ASK}(t)$ look like!?

ASK-Modulated Signal: Frequency Spectrum

$$\cos A \cdot \cos B = \frac{1}{2}(\cos(A-B)+\cos(A+B))$$

Carrier signal: $v_c(t) = \cos(2\pi f_c t) = \cos(\omega_c t)$, **where** $2\pi f_c = \omega_c$

Digital signal: **(unipolar!!!)**

$$v_d(t) = A \cdot \left[\frac{1}{2} + \frac{2}{\pi}\cos\omega_0 t - \frac{2}{3\pi}\cos 3\omega_0 t + \frac{2}{5\pi}\cos 5\omega_0 t - \ldots\right]$$

Modulated signal:

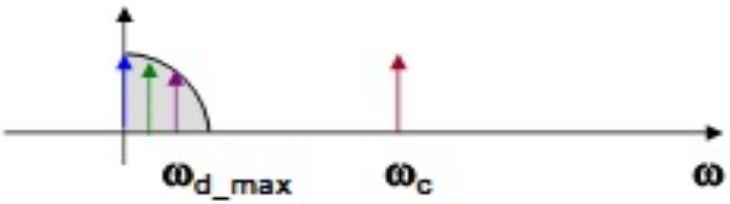
$$v_{ASK}(t) = v_c(t)\cdot v_d(t) =$$

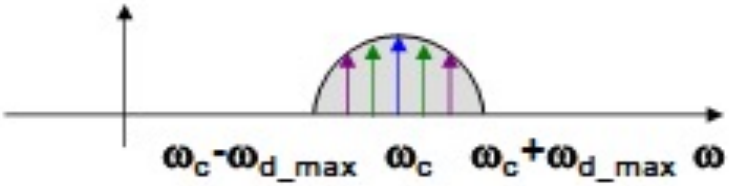
$$= \cos\omega_c t \cdot \left[\frac{1}{2} + \frac{2}{\pi}\cos\omega_0 t - \frac{2}{3\pi}\cos 3\omega_0 t + \frac{2}{5\pi}\cos 5\omega_0 t - \ldots\right] =$$

$$= \frac{1}{2}\cos\omega_c t + \frac{2}{\pi}\cos\omega_c t \cdot \cos\omega_0 t - \frac{2}{3\pi}\cos\omega_c t \cdot \cos 3\omega_0 t + \ldots =$$

$$= \frac{1}{2}\cos\omega_c t + \frac{1}{\pi}\left[\cos(\omega_c - \omega_0)t + \cos(\omega_c + \omega_0)t\right] -$$

$$- \frac{1}{3\pi}\left[\cos(\omega_c - 3\omega_0)t + \cos(\omega_c + 3\omega_0)t\right] + \ldots$$



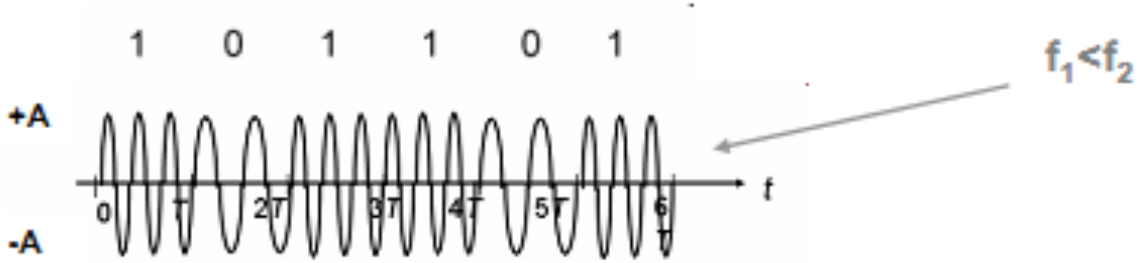


Modulation of Digital Data: FSK

FSK – frequency of carrier signal is varied to represent binary 1 or 0

- peak amplitude & phase remain constant during each bit interval

$$s(t) = \begin{cases} A\cos(2\pi f_1 t), & \text{binary 0} \\ A\cos(2\pi f_2 t), & \text{binary 1} \end{cases}$$

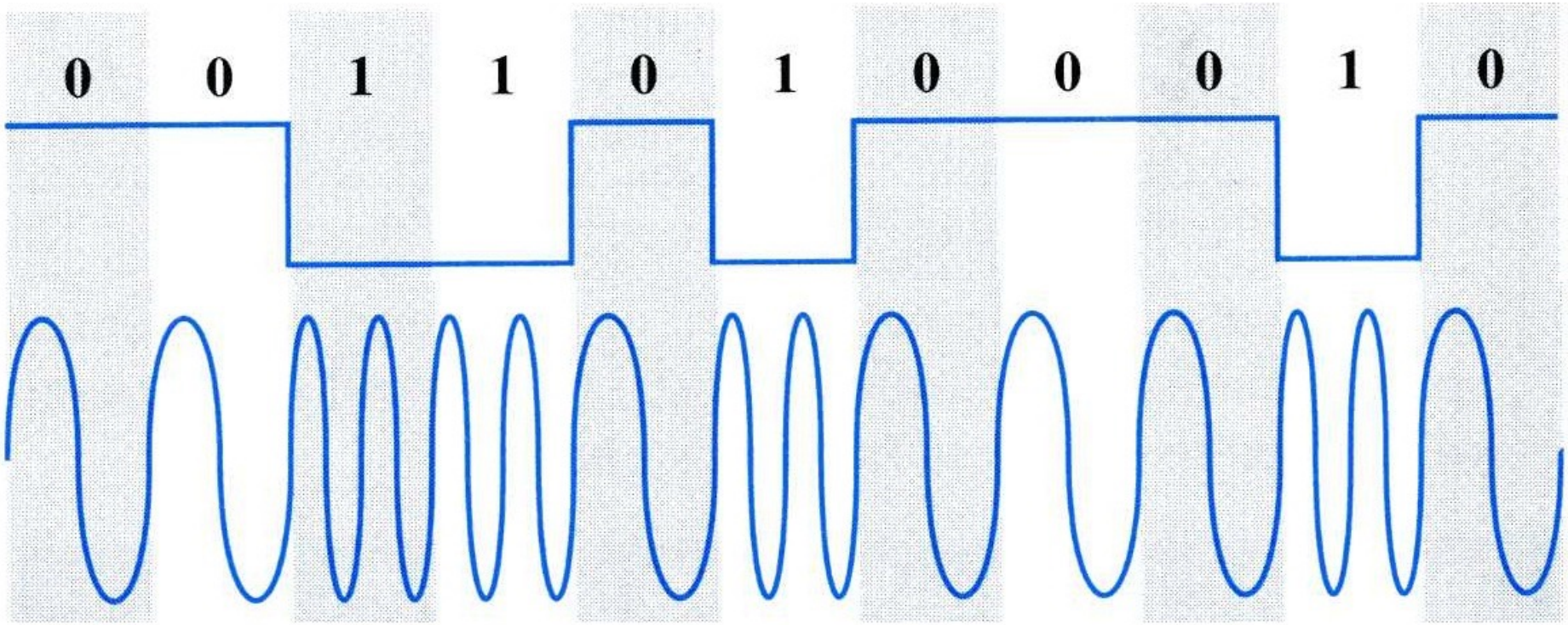


- demodulation: demodulator must be able to determine which of two possible frequencies is present at a given time
- advantage: FSK is less susceptible to errors than ASK – receiver looks for specific frequency changes over a number of intervals, so voltage (noise) spikes can be ignored
- disadvantage: FSK spectrum is 2 x ASK spectrum
- application: over voice lines, in high-freq. radio transmission, etc.

- Frequency Shift Keying (FSK)
  - The most basic (binary) form of FSK involves the process of varying the frequency of a carrier wave by choosing one of two frequencies (binary FSK) in correspondence to a sequence of digital pulses that constitute the information signal. Two binary digits are represented by two frequencies around the carrier frequency. Amplitude remains fixed

- FSK (Frequency Shift Keying): a simpler form ofFrequencyModulation for transmitting digital/binary data.
- FSK has better performance than ASK. Have no problem with sudden noise spikes.
- FSK was used in low speed (< 1200 bps) modems such as the venerable 300bps full-duplex.

- Secara umum berbentuk binary FSK (BFSK)
- Dua hasil biner diwakili oleh dua frekuensi yang berbeda (carrier dekat)
- Tidak mudah error daripada ASK
- Sampai dengan 1200bps pada voice grade line
- Frekuensi radio tinggi
- Tiap frekuen si tinggi pada LAN menggunakan koaksial

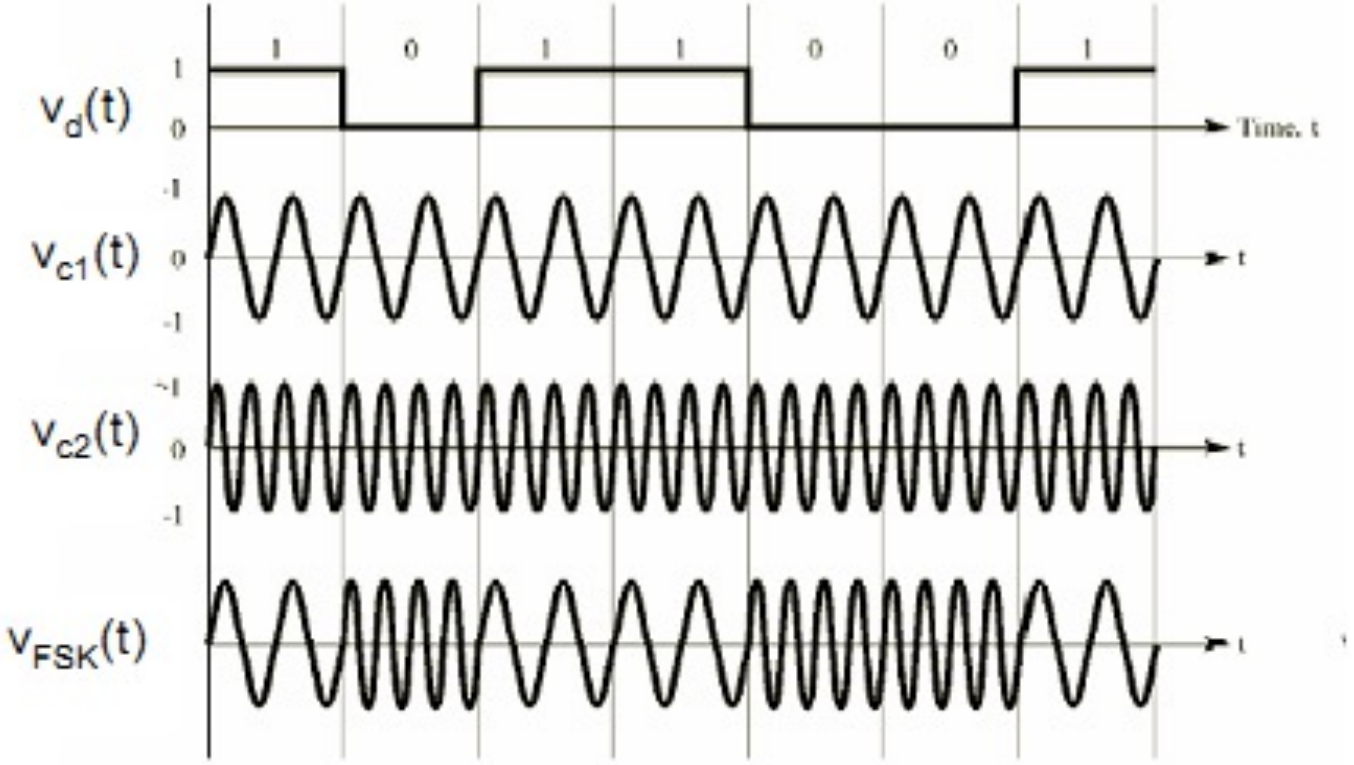


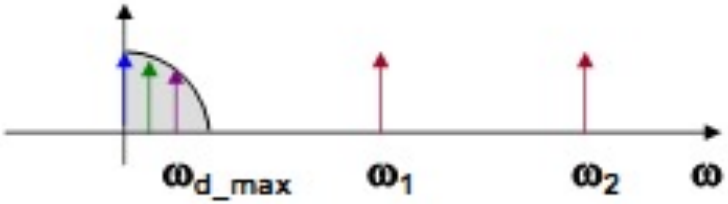
Frequency Shift Keying (FSK)

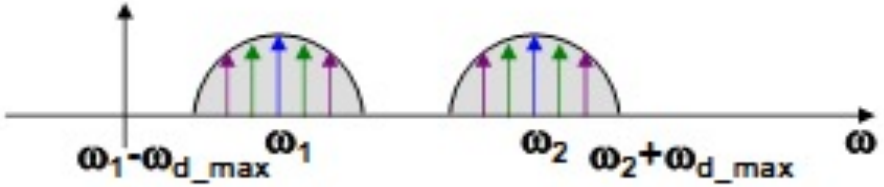
Two frequencies to represent 0 & 1

Modulation of Digital Data: FSK (cont.)

Example [FSK]







FSK-Modulated Signal: Frequency Spectrum

Digital signal: $v_d(t)$ **- modulated with** ω_1 **, and**

$v_d'(t) = 1 - v_d(t)$ **- modulated with** ω_2

Modulated signal:

$$v_{FSK}(t) = \cos\omega_1 t \cdot v_d(t) + \cos\omega_2 t \cdot (1 - v_d(t)) =$$

$$= \cos\omega_1 t \cdot \left[\frac{1}{2} + \frac{2}{\pi}\cos\omega_0 t - \frac{2}{3\pi}\cos 3\omega_0 t + \frac{2}{5\pi}\cos 5\omega_0 t - \ldots\right] +$$

$$+ \cos\omega_2 t \cdot \left[\frac{1}{2} - \frac{2}{\pi}\cos\omega_0 t + \frac{2}{3\pi}\cos 3\omega_0 t - \frac{2}{5\pi}\cos 5\omega_0 t - \ldots\right] =$$

$$= \ldots$$

$$= \frac{1}{2}\cos\omega_1 t + \frac{1}{\pi}\left[\cos(\omega_1 - \omega_0)t + \cos(\omega_1 + \omega_0)t\right] -$$

$$- \frac{1}{3\pi}\left[\cos(\omega_1 - 3\omega_0)t + \cos(\omega_1 + 3\omega_0)t\right] + \ldots +$$

$$\frac{1}{2}\cos\omega_2 t - \frac{1}{\pi}\left[\cos(\omega_2 - \omega_0)t + \cos(\omega_2 + \omega_0)t\right] -$$

$$+ \frac{1}{3\pi}\left[\cos(\omega_2 - 3\omega_0)t + \cos(\omega_2 + 3\omega_0)t\right] + \ldots +$$

Modulation of Digital Data: PSK

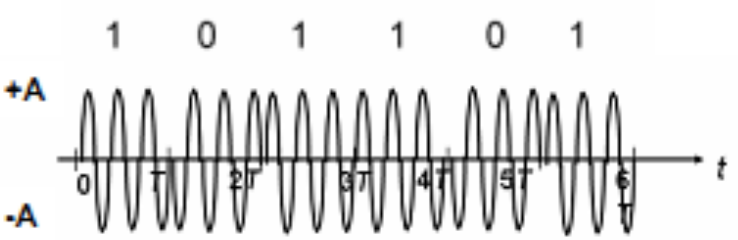
PSK – phase of carrier signal is varied to represent binary 1 or 0

- peak amplitude & freq. remain constant during each bit interval
- example: binary 1 = 0° phase, binary 0 = 180° (πrad) phase
  ⇒ PSK is equivalent to multiplying carrier signal by +1 when the information is 1, and by -1 when the information is 0

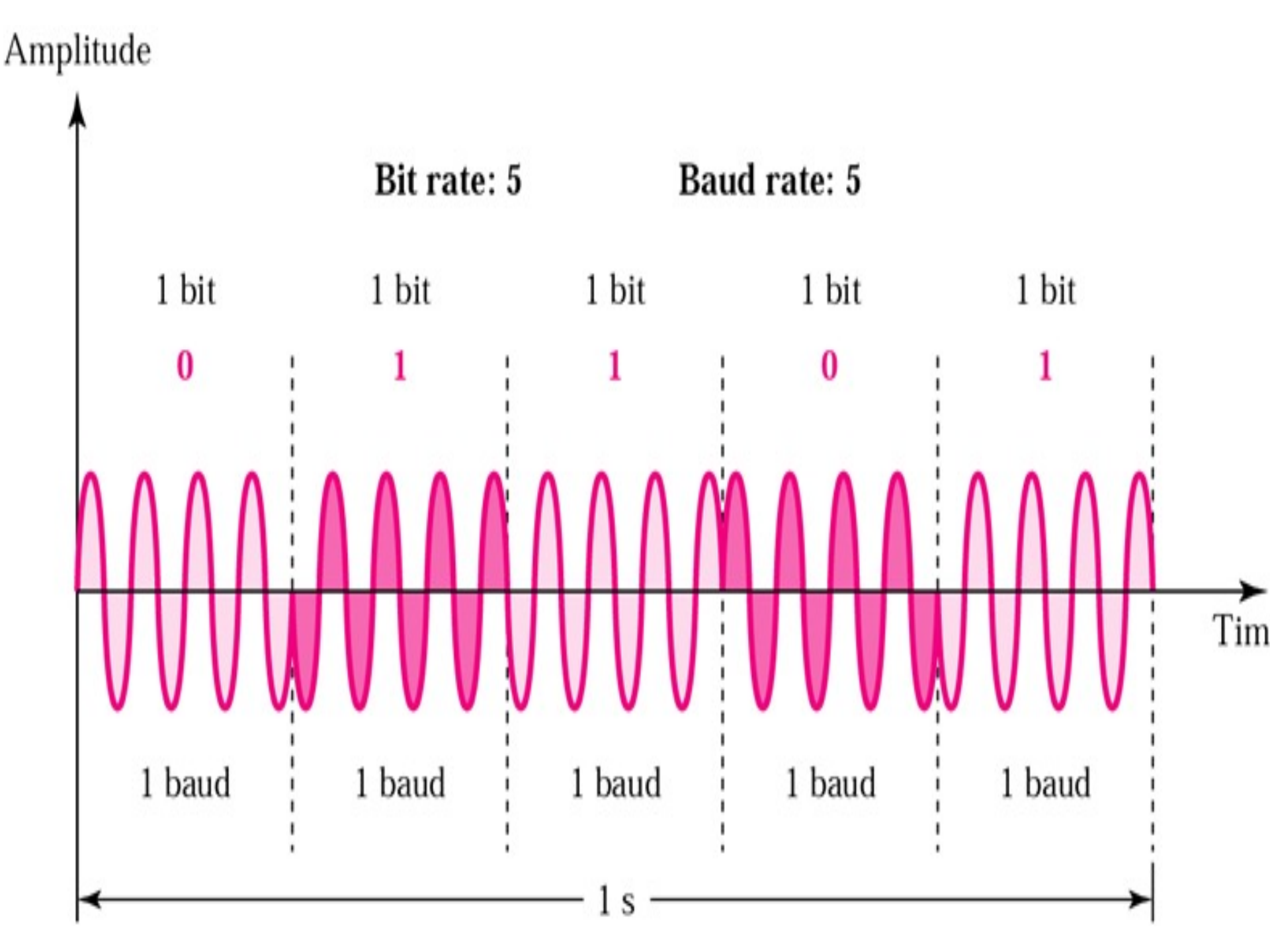
2-PSK, or **Binary PSK**, since only 2 different phases are used.

$$s(t) = \begin{cases} A\cos(2\pi f_c t), & \text{binary 1} \\ A\cos(2\pi f_c t + \pi), & \text{binary 0} \end{cases}$$

$$s(t) = \begin{cases} A\cos(2\pi f_c t), & \text{binary 1} \\ -A\cos(2\pi f_c t), & \text{binary 0} \end{cases}$$

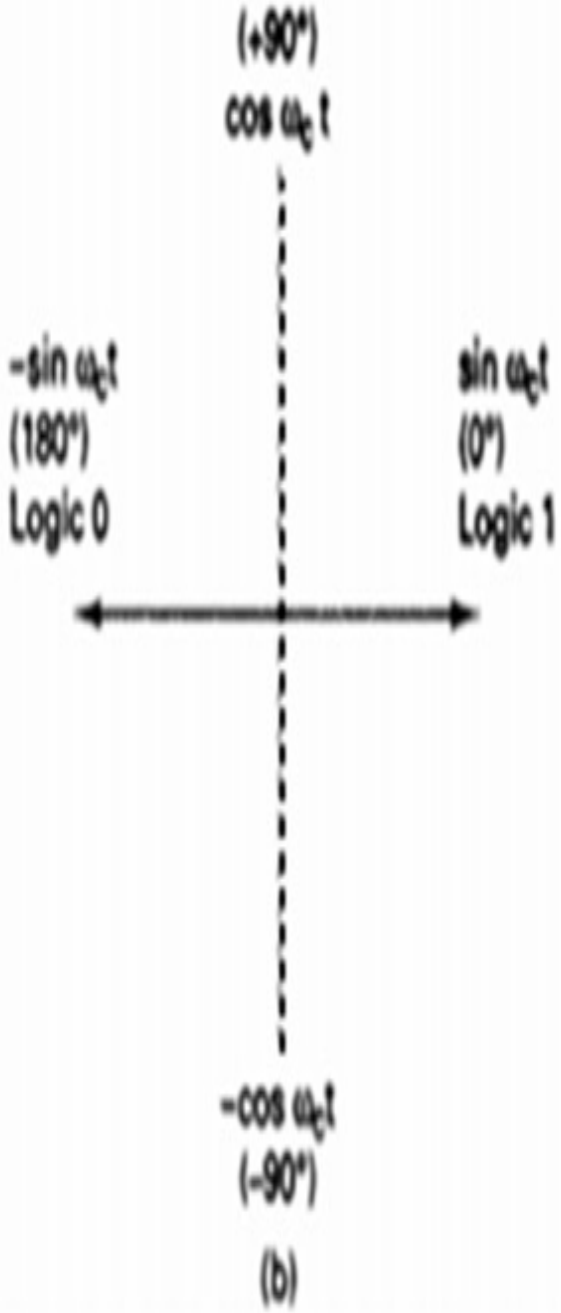


- demodulation: demodulator must determine the phase of received sinusoid with respect to some reference phase
- advantage:
  - PSK is less susceptible to errors than ASK, while it requires/occupies the same bandwidth as ASK
  - more efficient use of bandwidth (higher data-rate) are possible, compared to FSK !!!
- disadvantage: more complex signal detection / recovery process, than in ASK and FSK

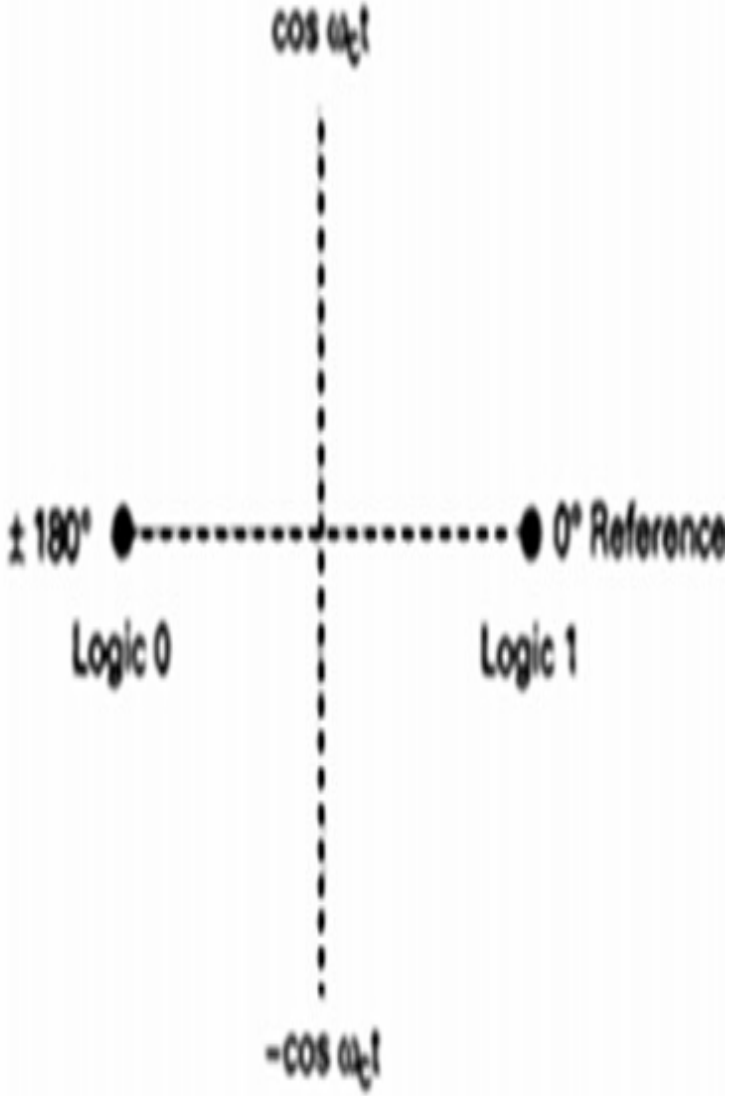

Amplitude
Bit rate: 5
Baud rate: 5
1 bit
1 bit
1 bit
1 bit
1 bit
0
1
1
0
1
Tim
1 baud
1 baud
1 baud
1 baud
1 baud
1 s

Binary input	Output phase
Logic 0	180°
Logic 1	0°

(a)



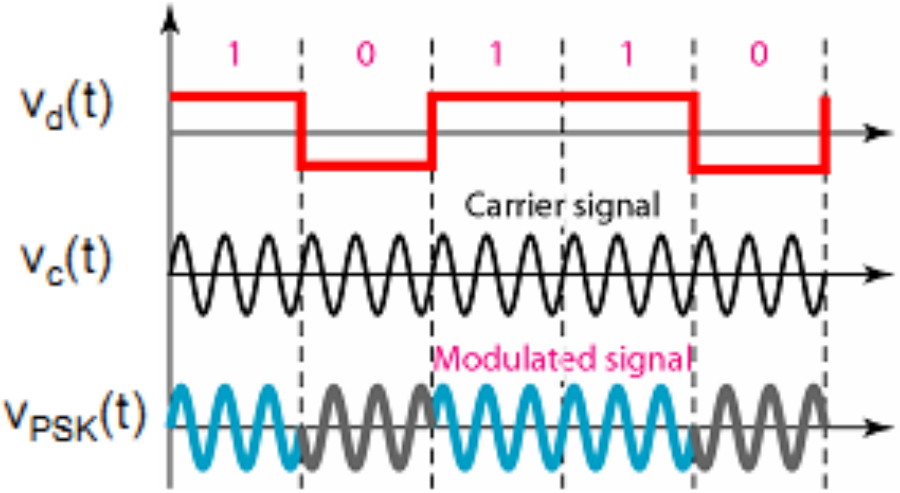
(b)

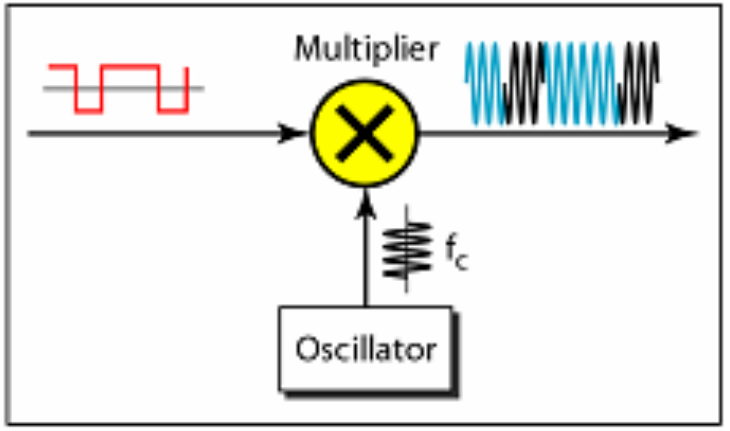


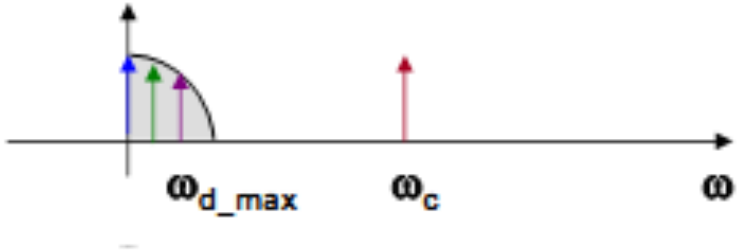
(c)

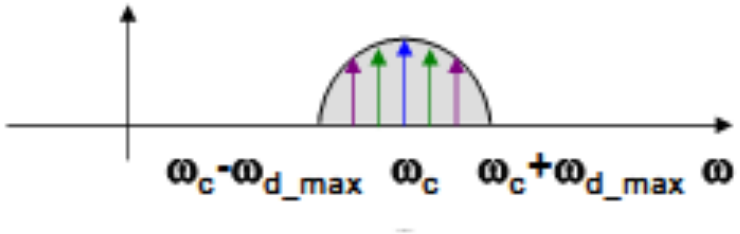
Modulation of Digital Data: PSK (cont.)

Example [PSK]









Modulation of Digital Data: PSK (cont.)

PSK Detection

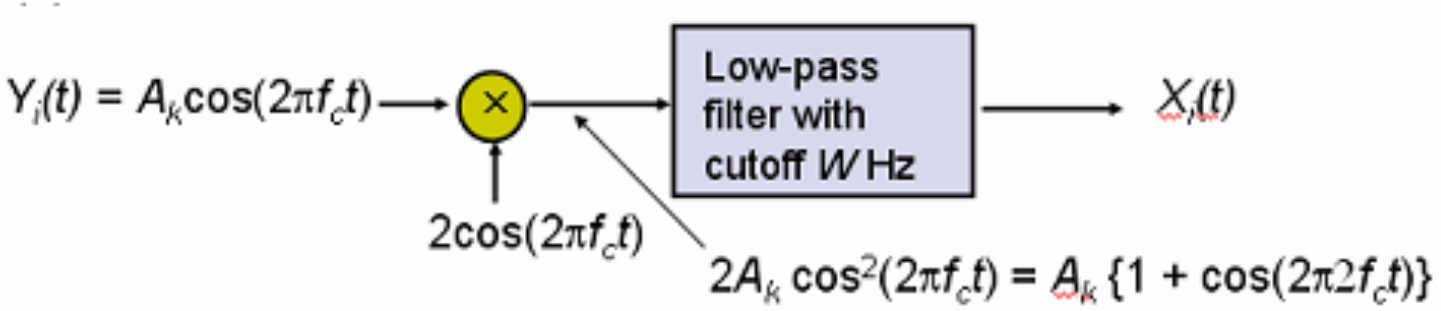
$$\cos^2 A = \frac{1}{2}(1+\cos 2A)$$

- **multiply the received / modulated signal** $\pm A\cos(2\pi f_c t)$ **by** $2*\cos(2\pi f_c t)$
  - **resulting signal**

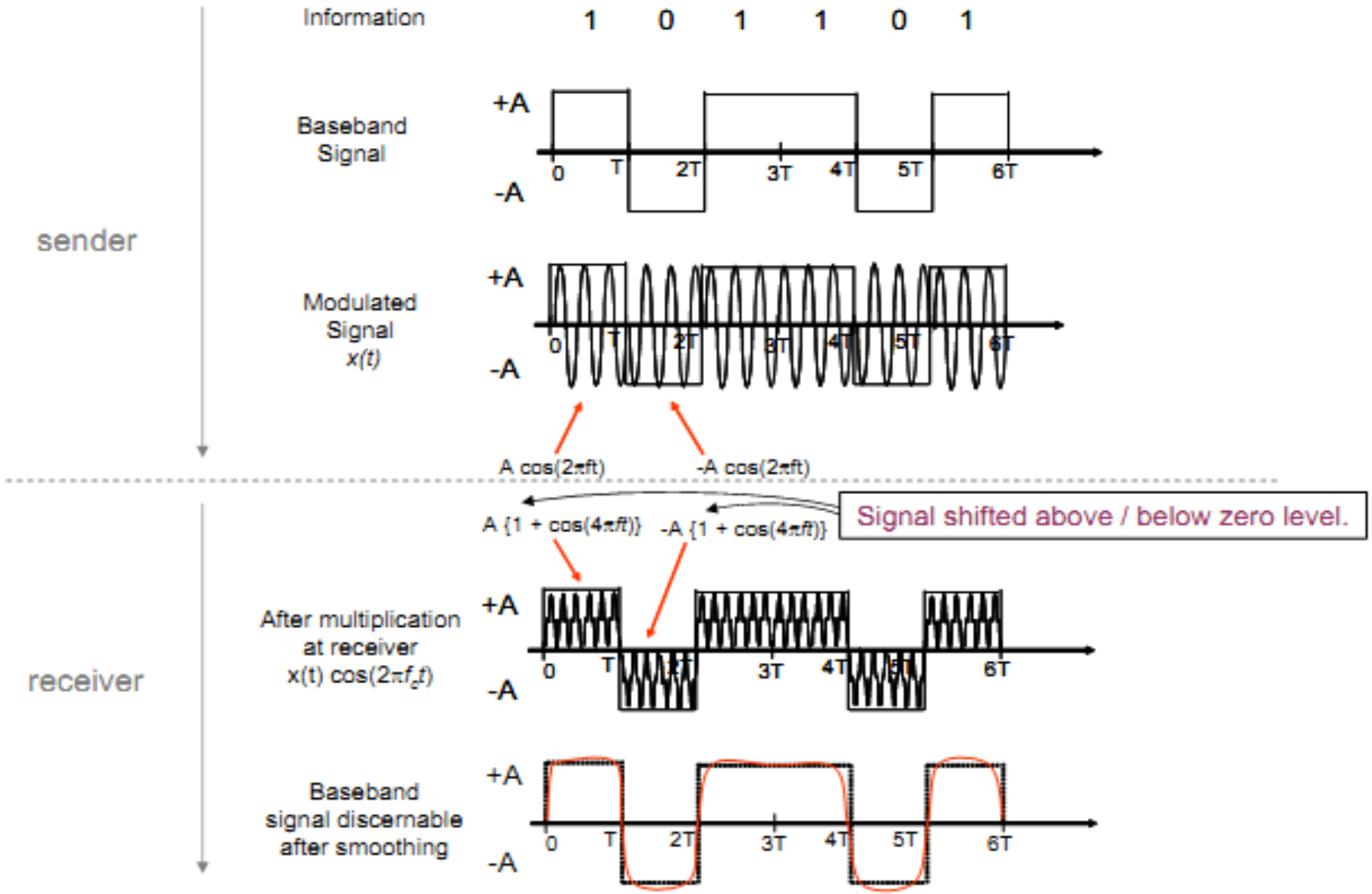
$$2A\cos^2(2\pi f_c t) = A[1+\cos(4\pi f_c t)], \quad \text{binary 1}$$

$$-2A\cos^2(2\pi f_c t) = -A[1+\cos(4\pi f_c t)], \quad \text{binary 0}$$

  - **by removing the oscillatory part with a low-pass filter, the original baseband signal (i.e. the original binary sequence) can be easily determined**



Modulation of Digital Data: PSK (cont.)



Modulation of Digital Data: PSK (cont.)

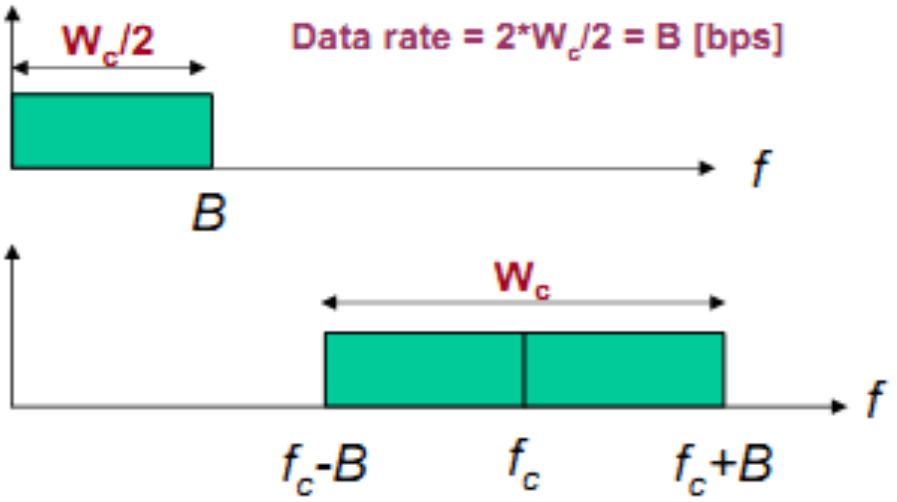
Facts from Modulation Theory

If

Baseband signal $x(t)$ with bandwidth $W_c/2$

then

Modulated signal $x(t)\cos(2\pi f_c t)$ has bandwidth W_c Hz



- If bandpass channel has bandwidth W_c [Hz],
  - then baseband channel has $W_c/2$ [Hz] available, so
  - modulation system supports $2*(W_c/2) = W_c$ [pulses/second]
  - recall Nyqyist Law: baseband transmission system of bandwidth W_c [Hz] can theoretically support 2 W_c pulses/sec

How can we recover the factor 2 in supported data-rate !?

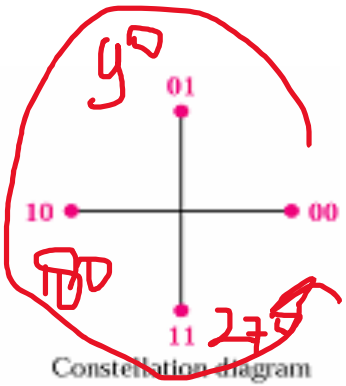
Modulation of Digital Data: PSK (cont.)

QPSK = 4-PSK – **PSK that uses phase shifts of 90°=π/2 rad ⇒ 4 different signals generated, each representing 2 bits**

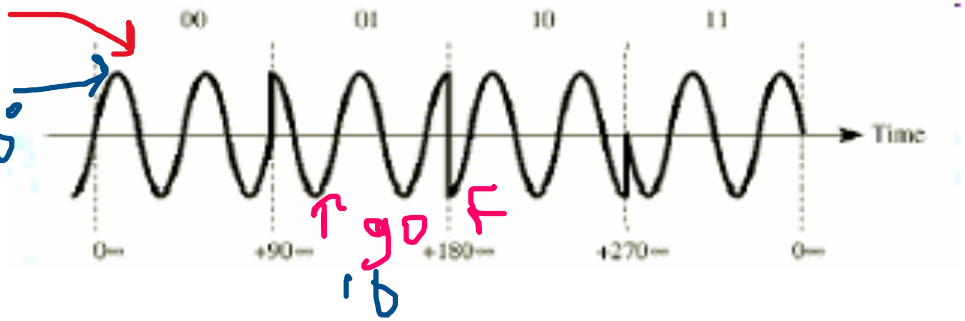
$$s(t) = \begin{cases} A\cos(2\pi f_c t), & \text{binary } 00 \\ A\cos(2\pi f_c t + \frac{\pi}{2}), & \text{binary } 01 \\ A\cos(2\pi f_c t + \pi), & \text{binary } 10 \\ A\cos(2\pi f_c t + \frac{3\pi}{2}), & \text{binary } 11 \end{cases}$$

Dibit	Phase
00	0
01	90
10	180
11	270

Dibit (2 bits)



Constellation diagram

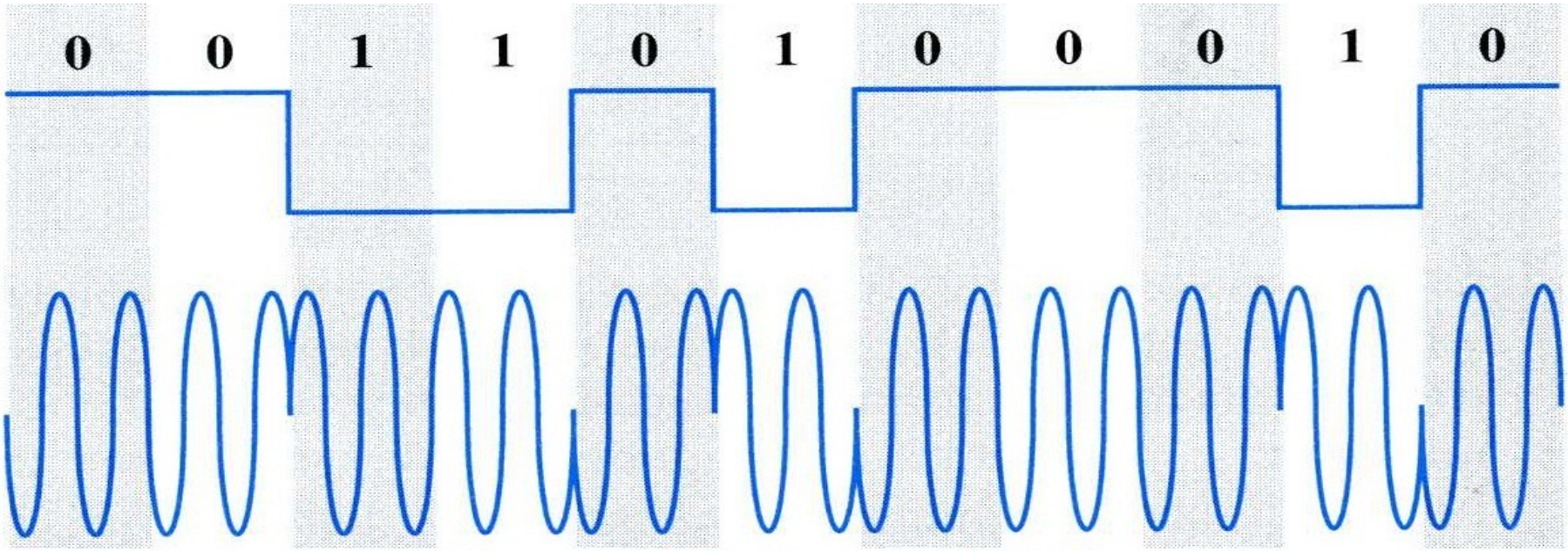


- **advantage**: higher data rate than in PSK (2 bits per bit interval), while bandwidth occupancy remains the same
- 4-PSK can easily be extended to 8-PSK, i.e. n-PSK
- however, higher rate PSK schemes are limited by the ability of equipment to distinguish small differences in phase

- PSK (Phase Shift Keying): a simpler form of Phase Modulation for transmitting digital/binary data.
- In PSK, the phase of the carrier signal is shifted (by 180°) to represent data.
- PSK variations: Bi-level PSK, Differential DPSK, and Quadrature QPSK.
- QPSK: 1 bps baud rate achieve 2 bps data rate.

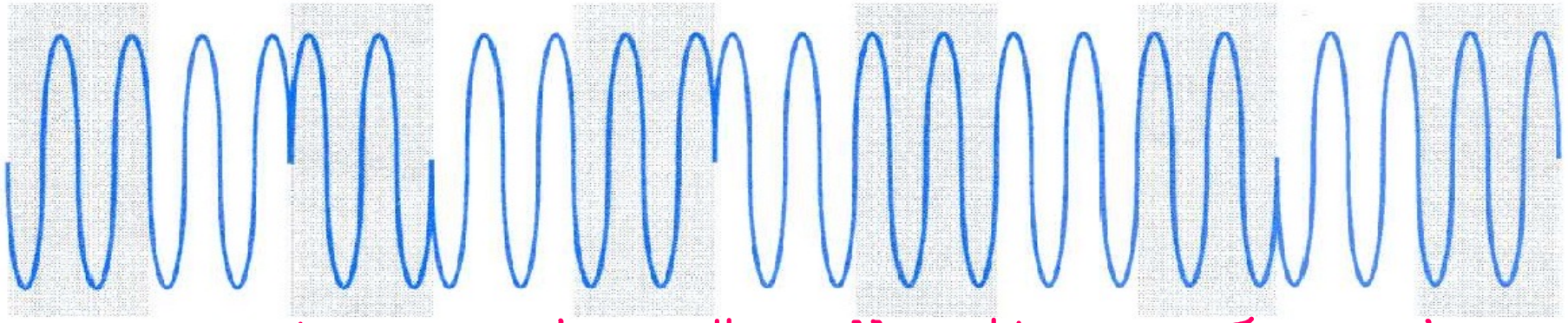
- Fase pada sinyal carrier adalah perubahan untuk mewakili data
- Binary PSK

  Dua fase diwakili dua digit biner
- Differential PSK
- Perubahan fase relatif untuk transmisi sebelumnya lebih dari beberapa sinyal referensi



Phase Shift Keying (PSK)

Or called BPSK, uses two phases to represent 0 & 1



Differential Phase Shift Keying (DPSK)

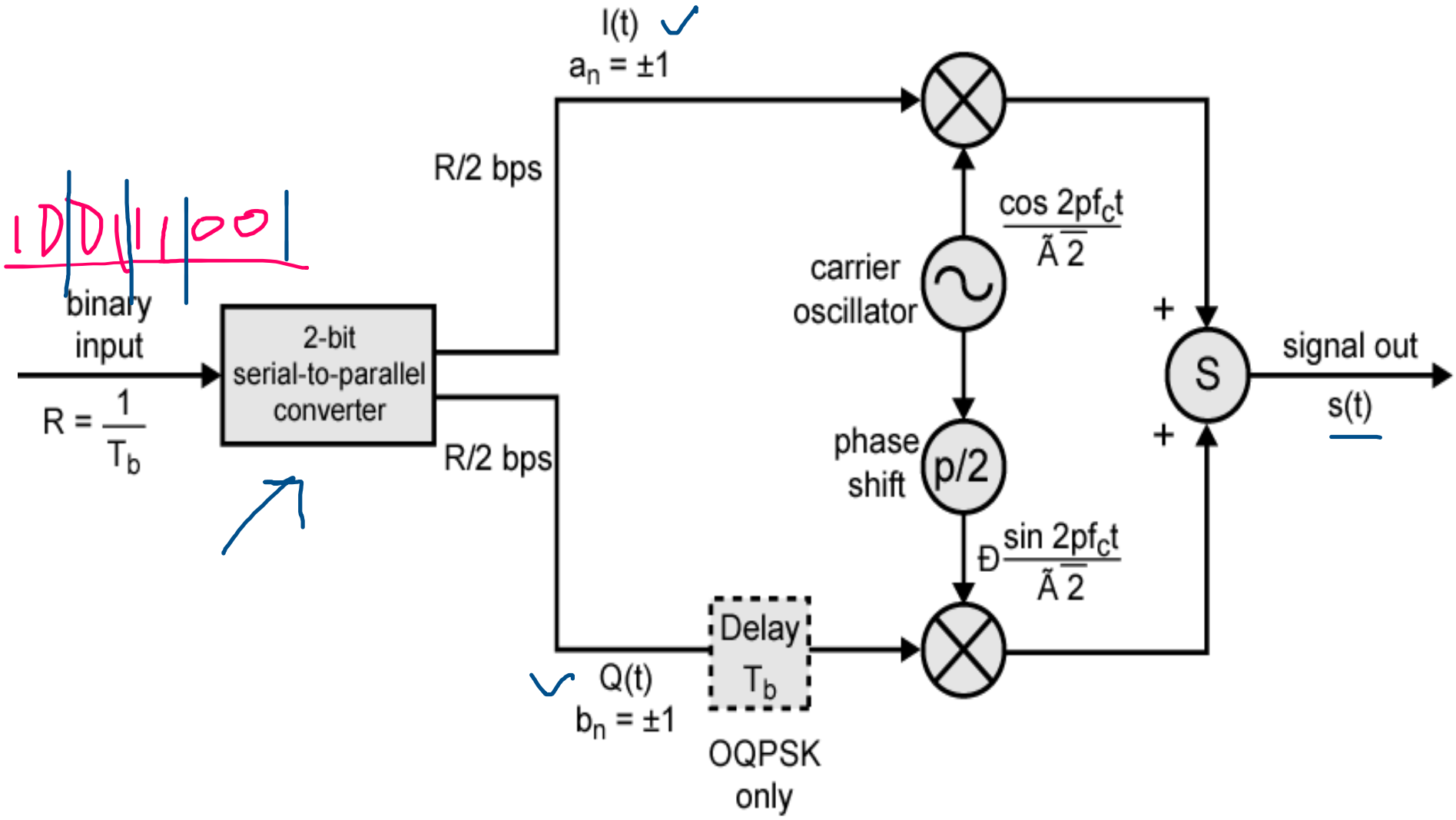
0 - no phase change

1 - phase change

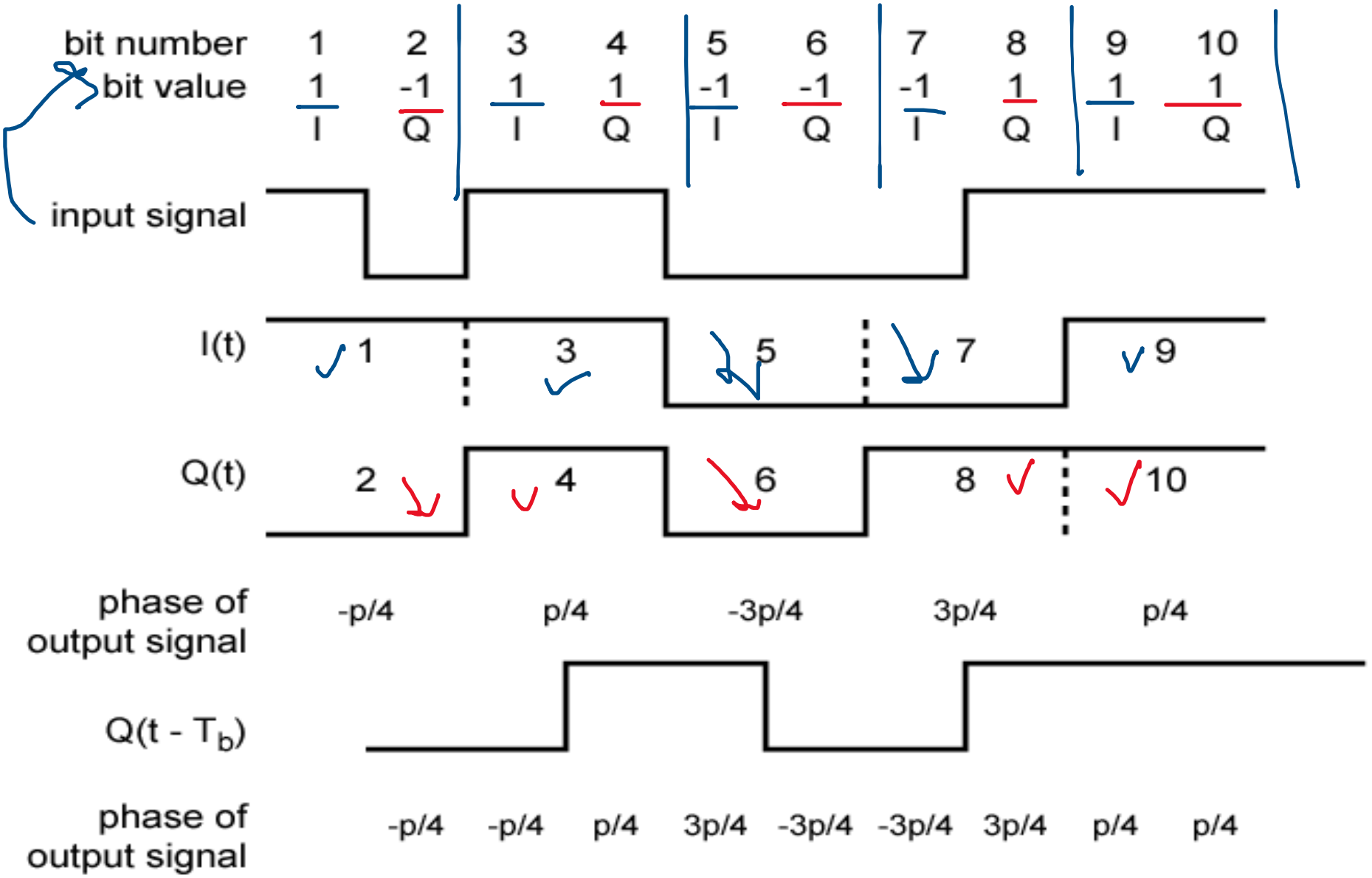
Quadrature PSK

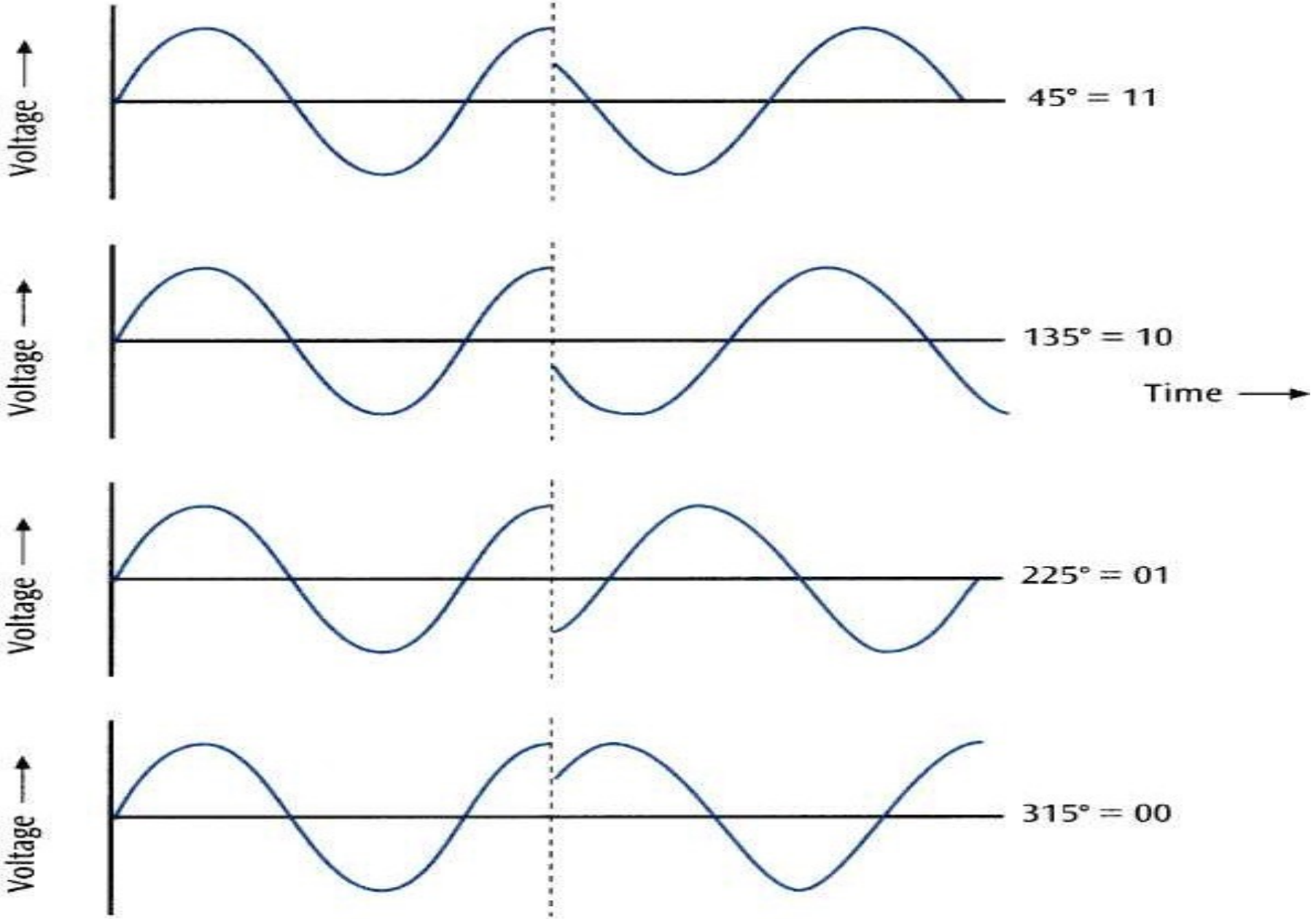
- Penggunaan lebih efisien oleh tiap elemen sinyal diwakili lebih dari satu bit
- Misalnya perubahan pada $\pi/2$ (90°)
- Tiap elemen diwakili dua bit
- Dapat digunakan 8 sudut fase dan memiliki lebih dari satu amplitudo
- 9600bps modem menggunakan sudut 12, empat pada tiap dua amplitudo
- Offset QPSK (orthogonal QPSK)
- Delay dalam aliran Q

QPSK dan Modulator OQPSK

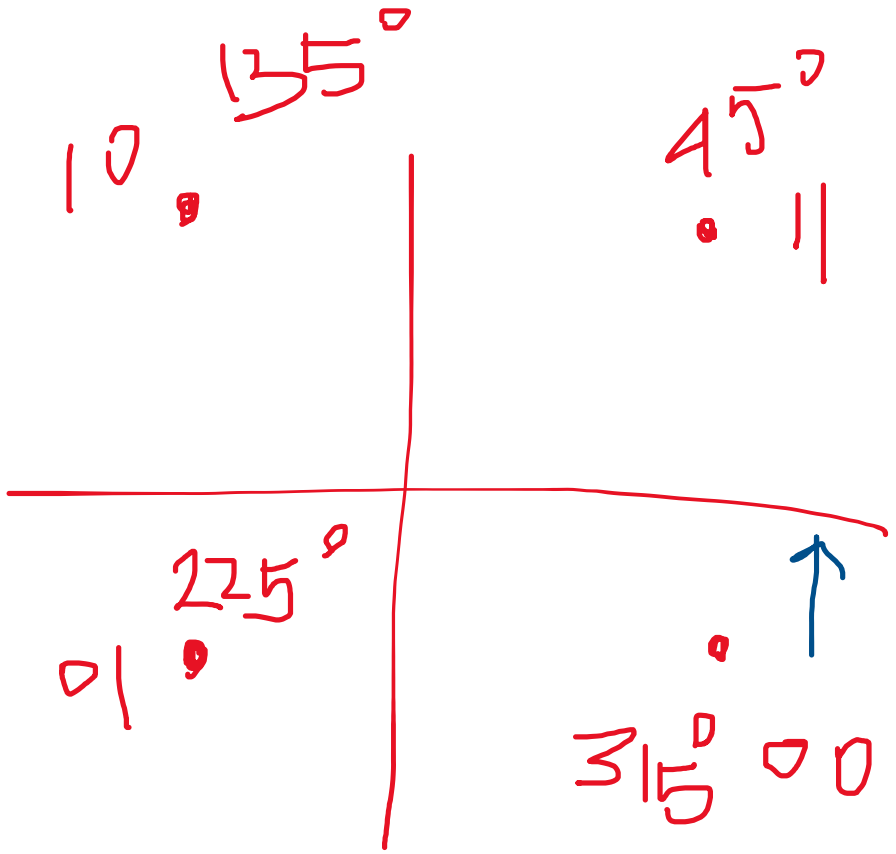


Contoh pada gelombang QPSK dan OQPSK





Quadrature PSK


135°
45°
225°
315°
BPSK

Modulation of Digital Data: QAM

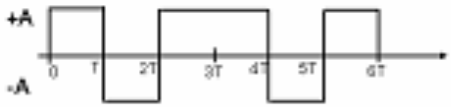
Quadrature Amplitude Modulation (QAM) – **uses "two-dimensional" signalling**

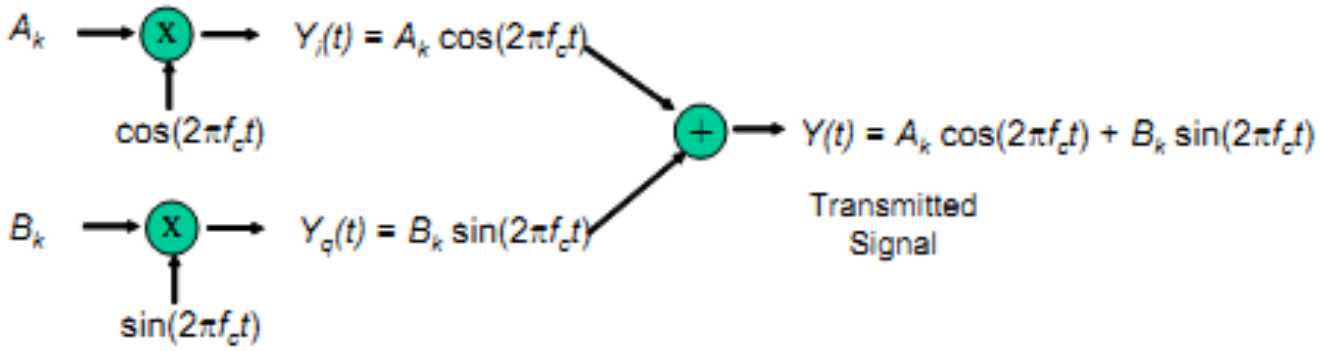
- **original information stream is split into two sequences that consist of odd and even symbols, e.g. B_k and A_k**

1	0	1	1	0	1	…
1	-1	1	1	-1	1	…
B_1	A_1	B_2	A_2	B_3	A_3	…

- **A_k sequence (in-phase comp.) is modulated by $\cos(2\pi f_c t)$**
  B_k sequence (quadrature-phase comp.) is modulated by $\sin(2\pi f_c t)$

- **composite signal $A_k \cos(2\pi f_c t) + B_k \sin(2\pi f_c t)$ is sent through the channel**

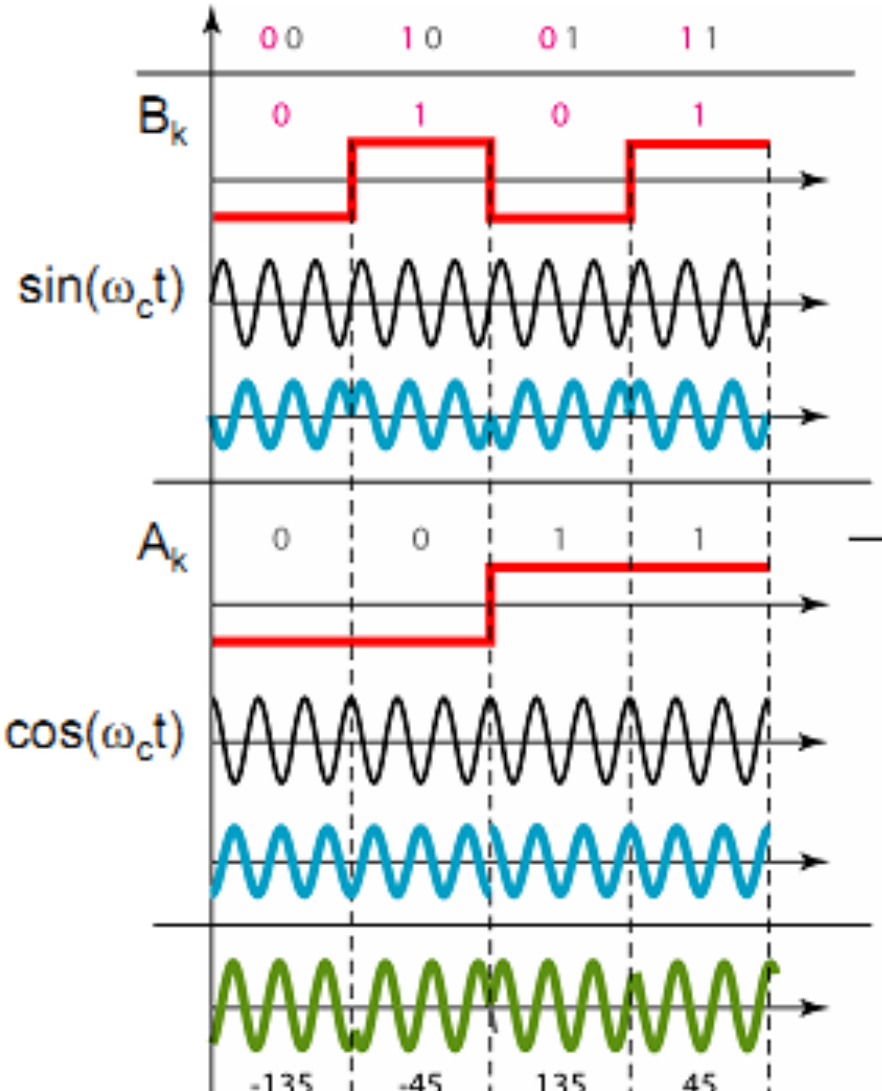


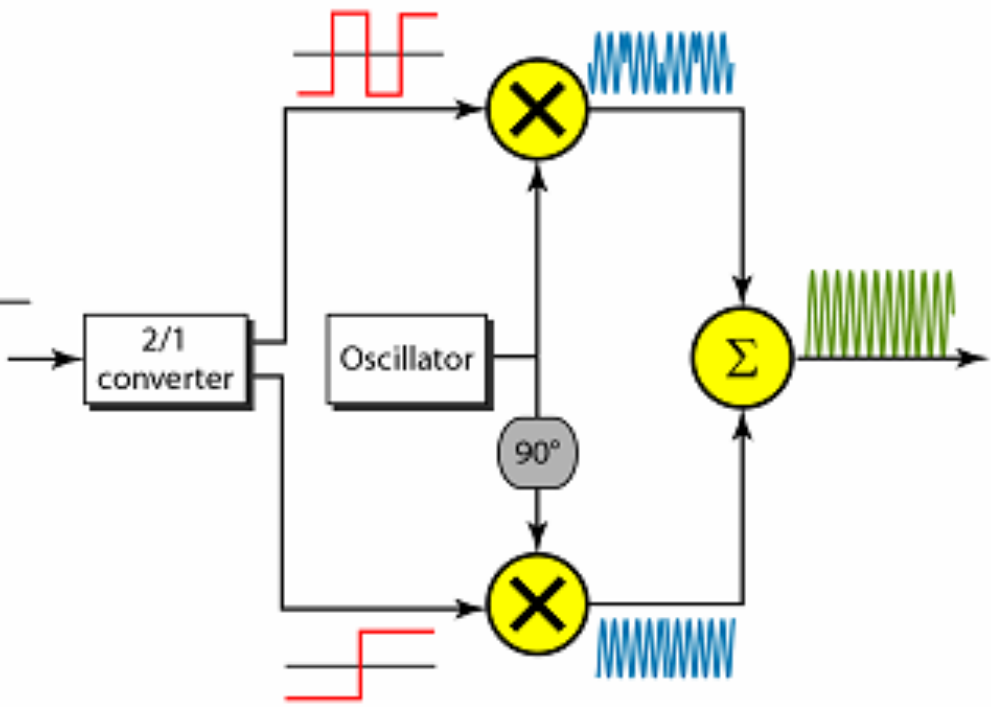


- **advantage: data rate = 2 bits per bit-interval!**

Modulation of Digital Data: QAM (cont.)

Example [QAM]

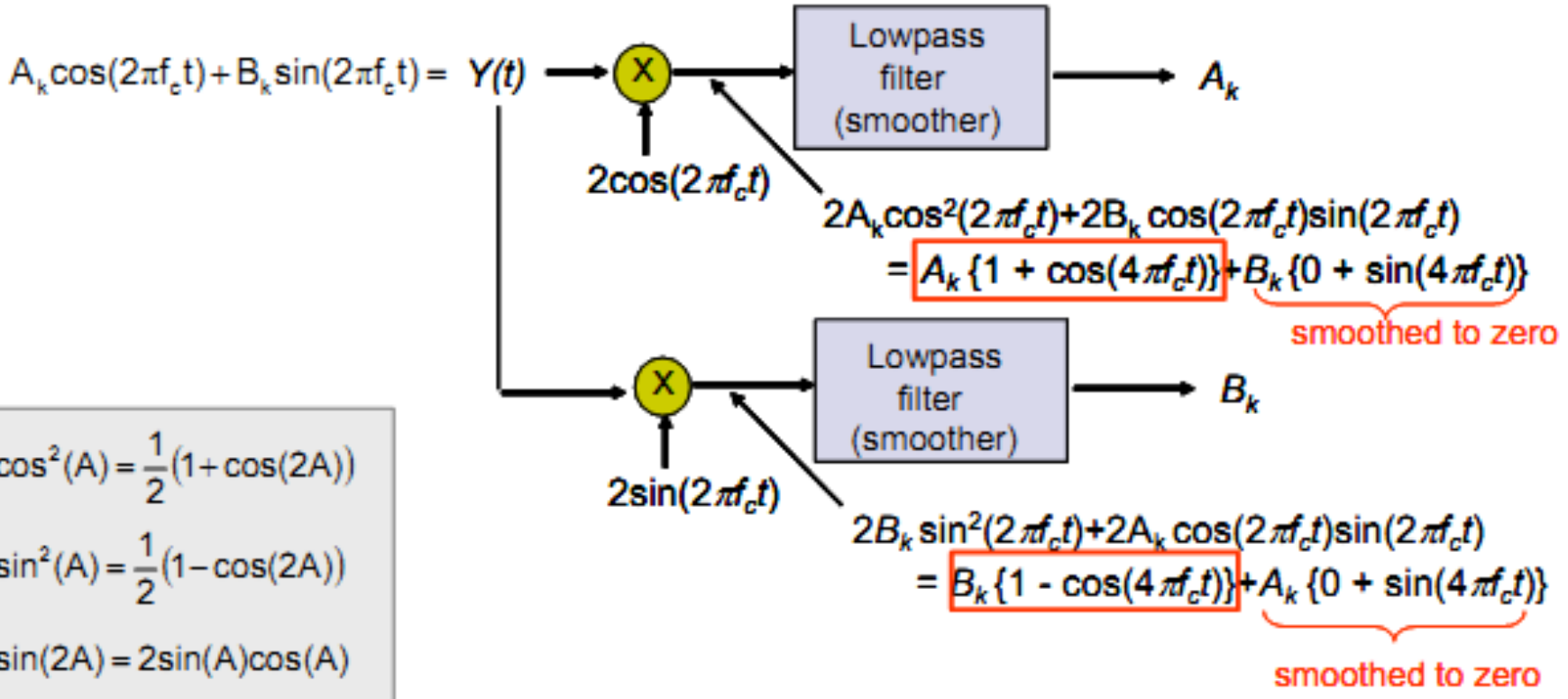




Modulation of Digital Data: QAM

QAM Demodulation

- by multiplying Y(t) by $2 \cdot \cos(2\pi f_c t)$ and then low-pass filtering the resultant signal, sequence A_k is obtained
- by multiplying Y(t) by $2 \cdot \sin(2\pi f_c t)$ and then low-pass filtering the resultant signal, sequence B_k is obtained



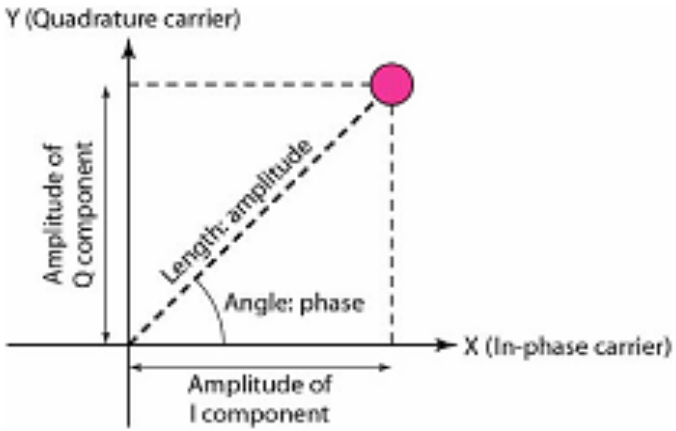
$$\cos^2(A) = \frac{1}{2}(1 + \cos(2A))$$

$$\sin^2(A) = \frac{1}{2}(1 - \cos(2A))$$

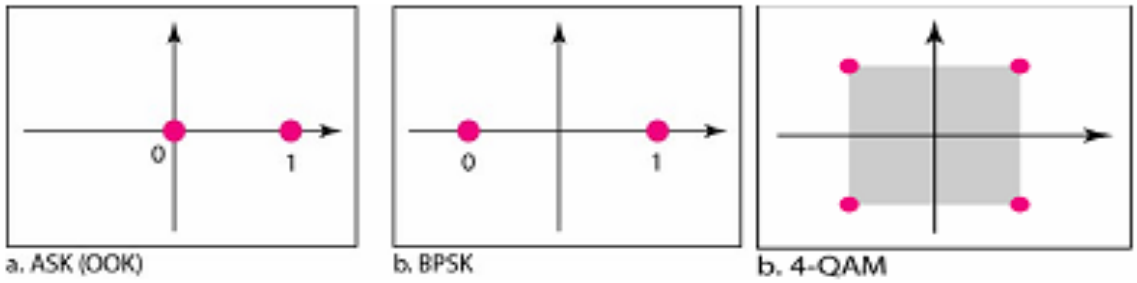
$$\sin(2A) = 2\sin(A)\cos(A)$$

Signal Constellation

Constellation Diagram – used to represents possible symbols that may be selected by a given modulation scheme as <u>points</u> in 2-D plane



- X-axis is related to in-phase carrier: $\cos(\omega_c t)$
  - the projection of the point on the X-axis defines the peak amplitude of the in-phase component
- Y-axis is related to quadrature carrier: $\sin(\omega_c t)$
  - the projection of the point on the Y-axis defines the peak amplitude of the quadrature component
- the length of line that connects the point to the origin is the peak amplitude of the signal element (combination of X & Y components)
- the angle the line makes with the X-axis is the phase of the signal element

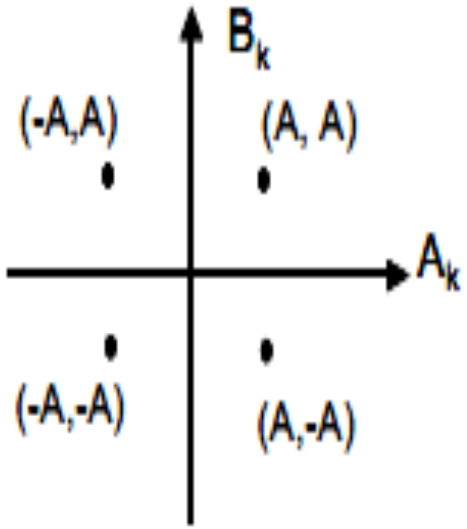


Modulation of Digital Data: QAM

QAM cont. – **QAM can also be seen as a combination of ASK & PSK**

$$Y(t) = A_k \cos(2\pi f_c t) + B_k \sin(2\pi f_c t) = \left(A_k^{\,2} + B_k^{\,2}\right)^{\frac{1}{2}} \cos\left(2\pi f_c t + \tan^{-1}\frac{B_k}{A_k}\right)$$

4-level QAM

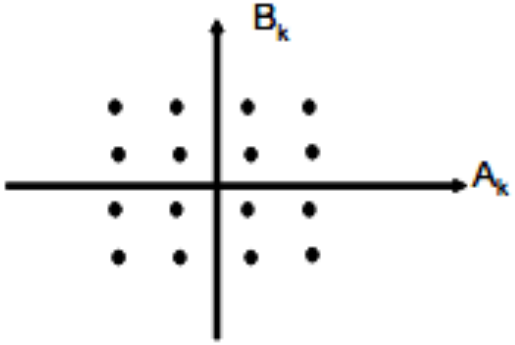


Modulation of Digital Data: QAM

16-level QAM – **the number of bits transmitted per T [sec] interval can be further increased by increasing the number of levels used**

- **in case of 16-level QAM, A_k and B_k individually can assume 4 different levels: -1, -1/3, 1/3, 1**
- **data rate: 4 bits/pulse ⇒ 4W bits/second**

$$Y(t) = A_k \cos(2\pi f_c t) + B_k \sin(2\pi f_c t) = \left(A_k^{\,2} + B_k^{\,2}\right)^{\frac{1}{2}} \cos\left(2\pi f_c t + \tan^{-1}\frac{B_k}{A_k}\right)$$



A_k and B_k individually can take on 4 different values; the resultant signal can take on (only) 3 different values!!!

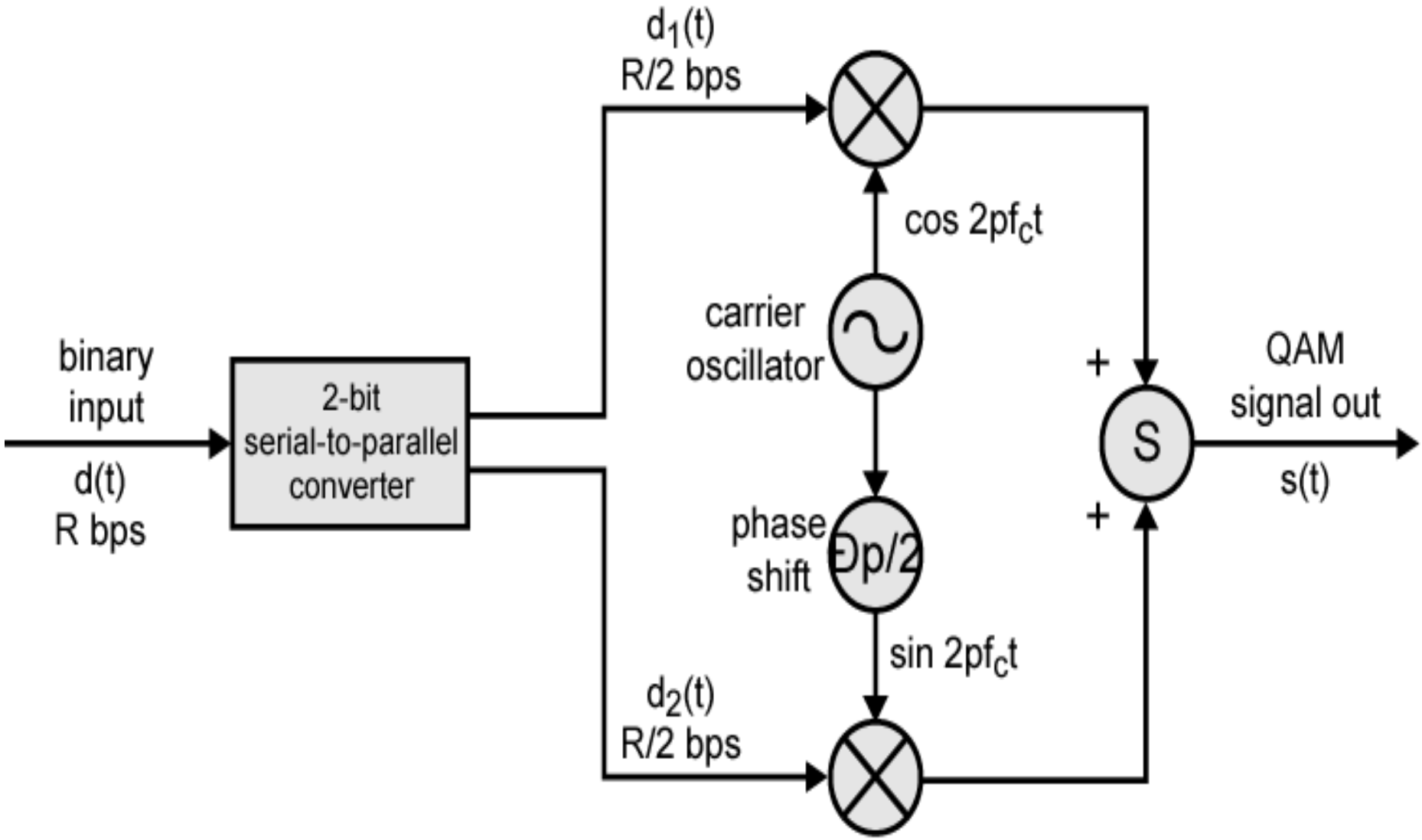
In QAM various combinations of amplitude and phase are employed to achieve higher digital data rates.

Amplitude changes are susceptible to noise ⇒ the number of phase shifts used by a QAM system is always greater than the number of amplitude shifts.

Quadrature Amplitude Modulation

- QAM digunakan pada asymmetric digital subscriber line (ADSL) dan beberapa wireless
- Kombinasi dari ASK dan PSK
- Logical extension pada QPSK
- Dikirimkan dua sinyal simultan yang berbeda dalam frekuensi carrier yang sama
- Digunakan dua copy carrier,satu shifted 90˚
- Tiap carrier adalah modulasi ASK
- Dua sinyal independen sama media
- Demodulasi dan kombinasi untuk output sinyal original

QAM Modulator



Sekian terima kasih, minggu depan UTS

Materi dari awal kuliah semester ini sampai akhir kuliah hari ini,
Thanks so much.....

DON'T BE SAD, AND DON'T BE ANGRY WITH ME.... OK